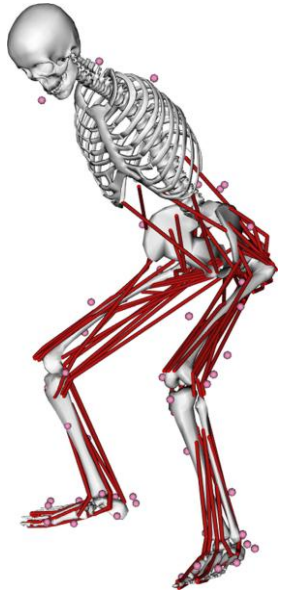
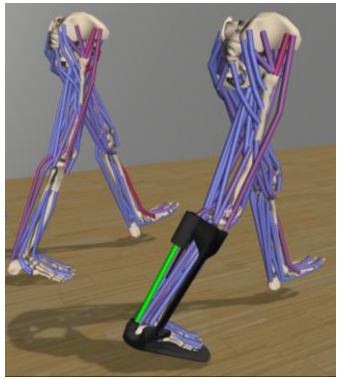
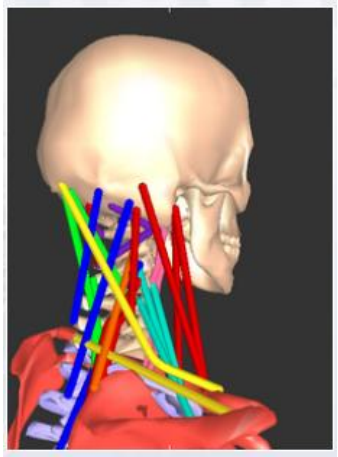
SM2501 Biomechanics of Human Movement: Computer Labs Intro

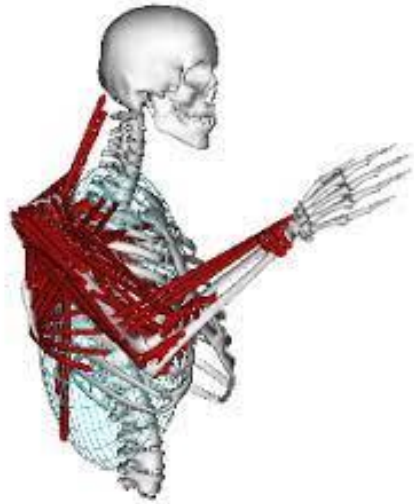
Ruoli Wang
Promobilia
MoveAbility Lab









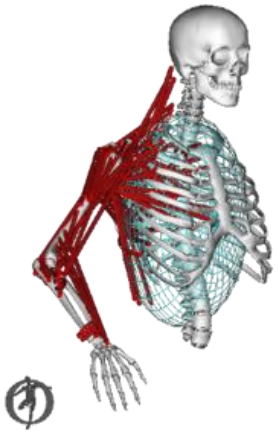


https://simtk.org/projects/opensim

Software: OpenSim v4.5 (@stanford.edu)

Promobilia MoveAbility Lab

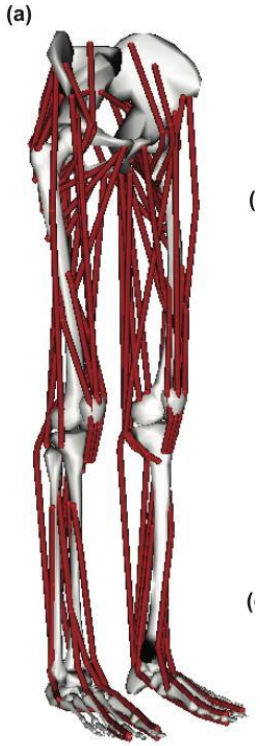
OpenSim



Seth A et al., Dynamic shoulder model



Christophy M et al., Lumbar spinal model

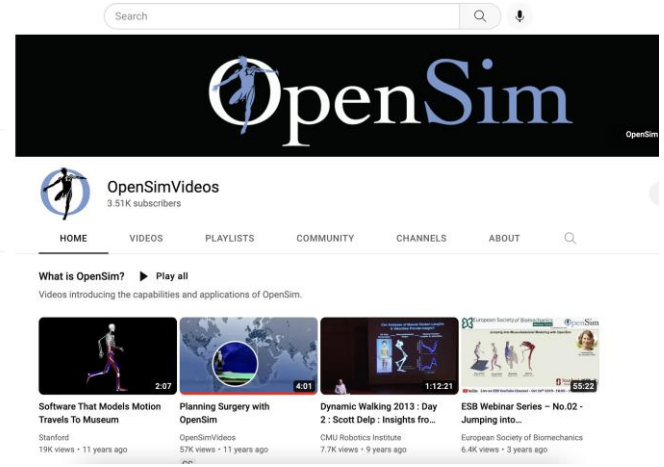


Rajagopal et al., Full-body Model

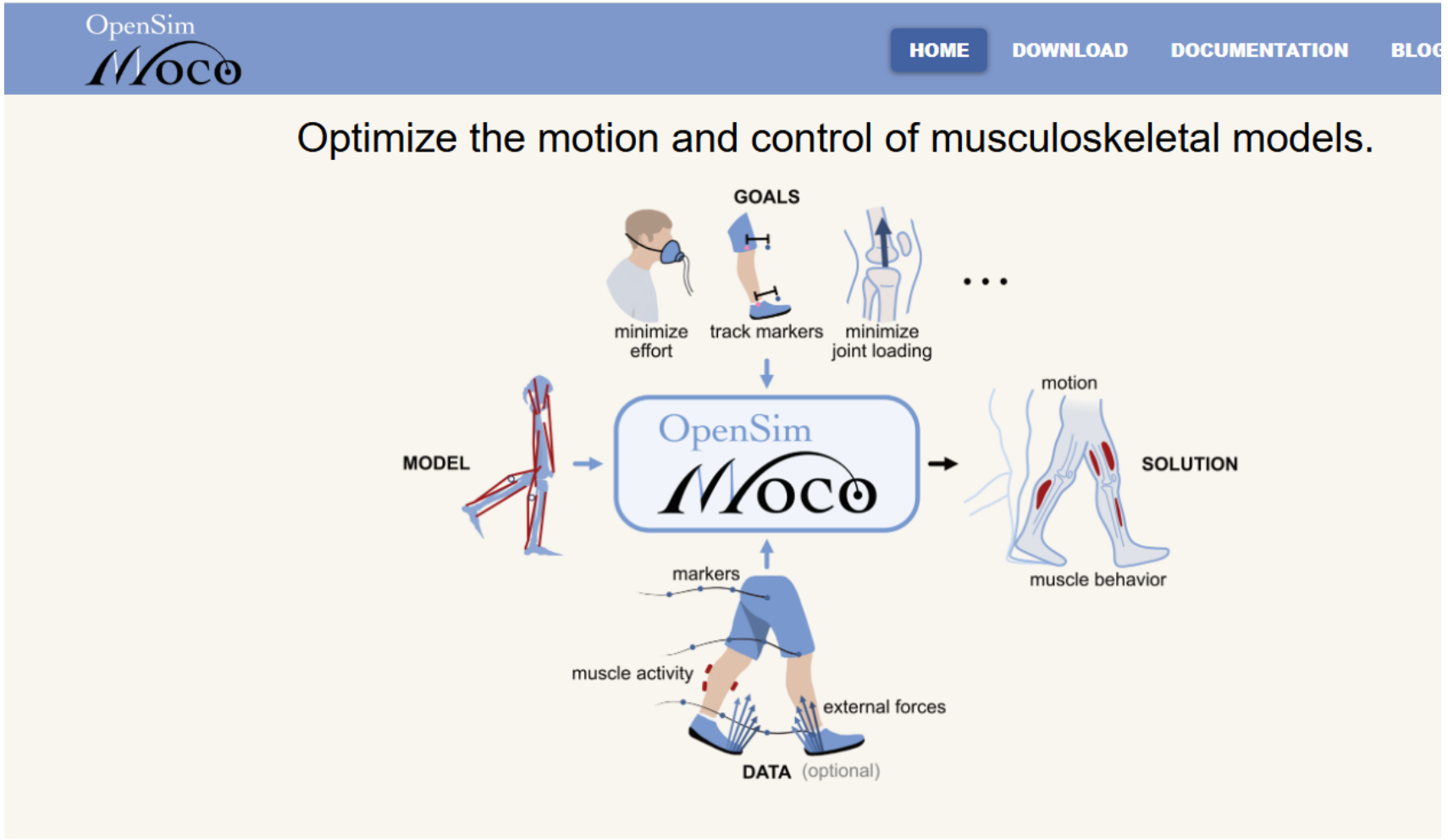
A software to create and analyze dynamic simulation of human movement

- **A repository of models, data & tools**
- **Analyze and simulate models and motions**
- **Customize workflows and extend the software**

https://www.youtube.com/user/opensimvideos



Software: OpenSim v4.5 (@stanford.edu)



https://opensim-org.github.io/opensim-moco-site/

Computer Lab Schedule HT 2025

		11.15		Backe 18, 4th fl (compulsory)		
45	Mon 3-Nov	10-12	Christopher	*Computer Lab 1*: Tutorial 1		*Canvas quiz*
	Tues 4-Nov	10-12	U21	L4: Inverse dynamics	Ch. 8	
	Wed 5-Nov	14-16	U21	L5: Muscle biomechanics and physiology I (R Wang)	Ch. 5-6	
		16-18	*MoveAbility*	*TA Office Hours*		
46	Mon 10-Nov	10-12	U21	L6: Muscle biomechanics II	Ch. 5-6	
	Tues 11-Nov	10-12	U21	L7: Kinematics of normal walking	Ch. 2	
		13-15	*MoveAbility*	*TA Office Hours (J Maier)*		
	Thurs 13-Nov	13-15	U21	L8: Kinetics and muscle activities in walking	Ch. 11	
		15-17	Christopher/ zoom	*Computer Lab 2*: OpenSIM IK + ID		*Canvas quiz*
	Fri 14-Nov					*Proj1 (Motion biomechanics) due 12.00*
47	Mon 17-Nov	10-12	Christopher/ zoom	*Computer Lab 3*: OpenSIM RRA + Static Optimization	Ch. 9-10	*Canvas quiz*
	Tues 18-Nov	10-12	U21	L9: Muscles III		
		13-15	*MoveAbility*	*TA Office Hours (J Maier)*		
	Thur	13-15	U21	L10: Motor control and muscle		

Computer lab1: **IRL, Christopher & Nils**

Computer lab2 and lab3: streaming

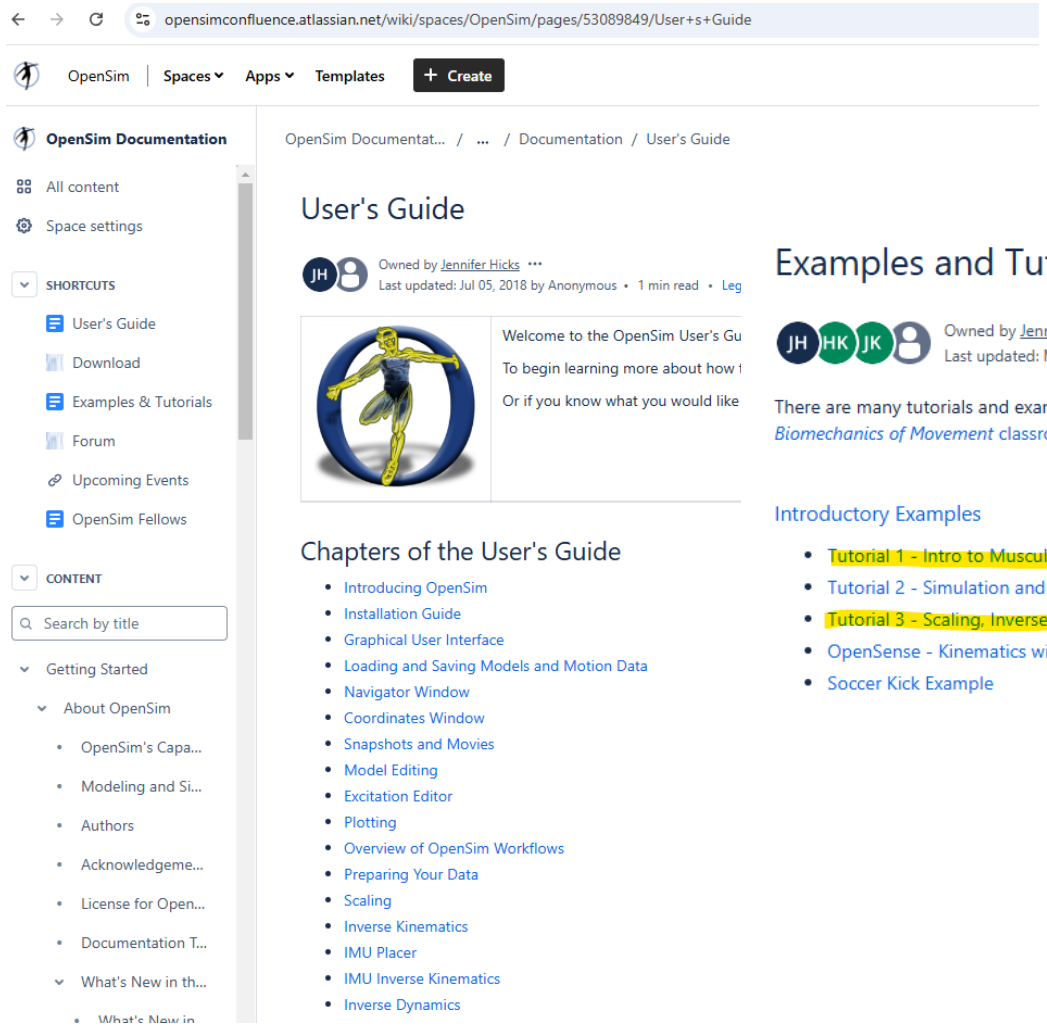
https://kth-se.zoom.us/j/7600710202

SM2501/FSG3084 HT25 › Assignments

Quiz OpenSim Computer Lab **Obligatory!**

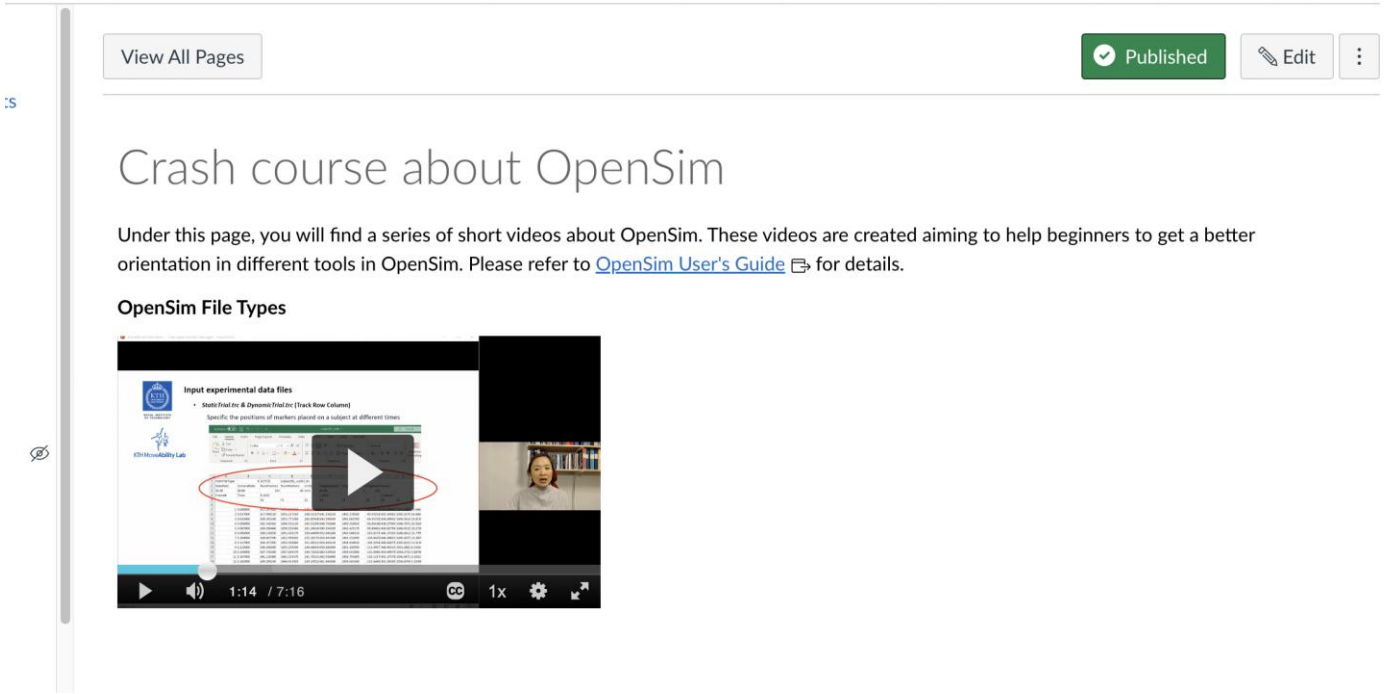
- Tutorial 1: Intro to Musculoskeletal Modeling
  Due 6 Nov at 19:00
- Tutorial 3: Scaling, Inverse Kinematics, and Inverse Dynamics
  Due 17 Nov at 19:00
- Working with static optimisation
  Due 20 Nov at 19:00

Tutorials



https://opensimconfluence.atlassian.net/wiki/spaces/OpenSim/overview

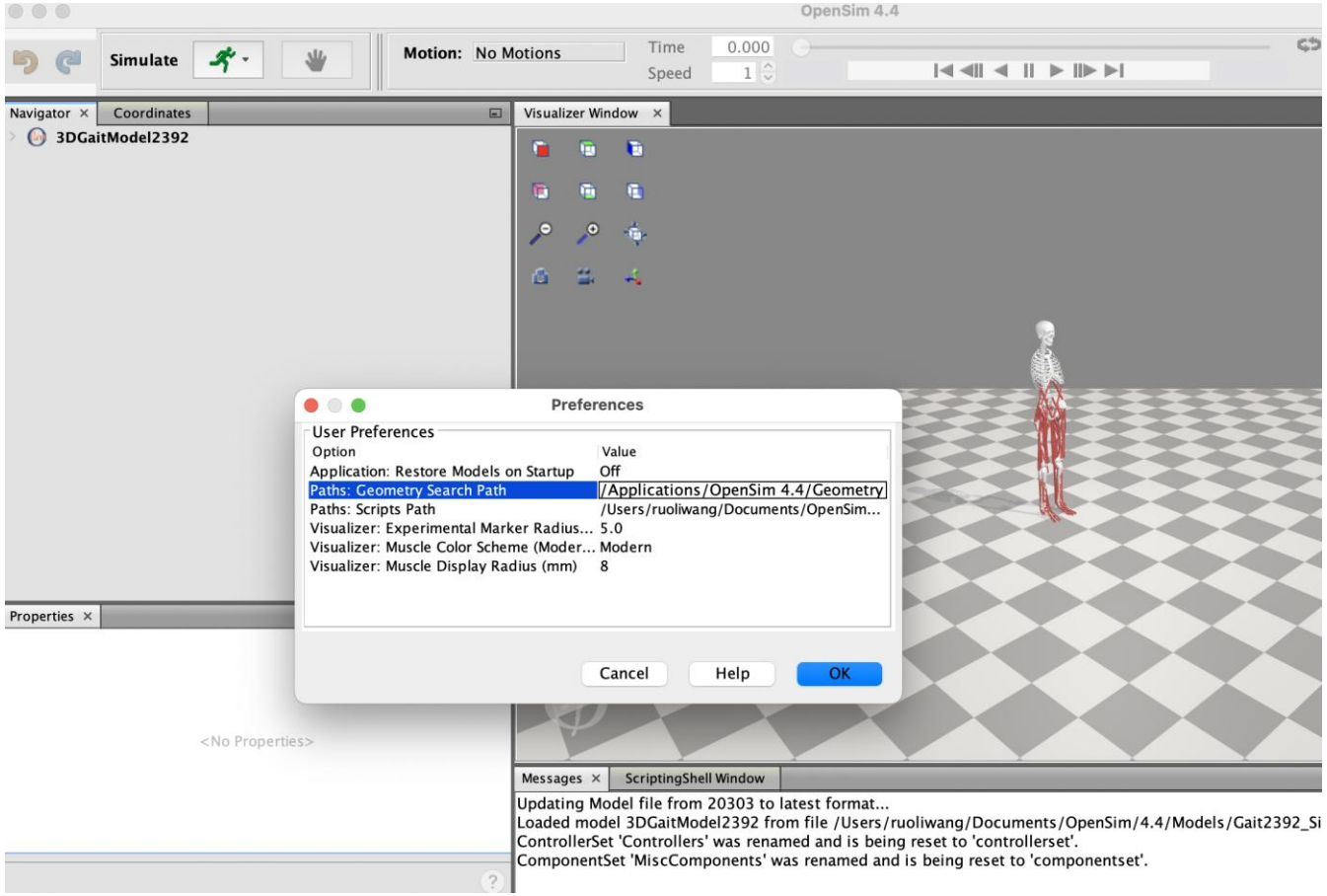
Crash course about OpenSim in Canvas



OpenSim File Types
OpenSim Workflow and Scale Tool
OpenSim Inverse Kinematics Tool
OpenSim Inverse Dynamics Tool
OpenSim Residual Reduction Tool
OpenSim Static Optimiaztion/CMC Tool

OpenSim Installation

Download from: https://simtk.org/frs/index.php?group_id=91



Correct Geometry Search Path

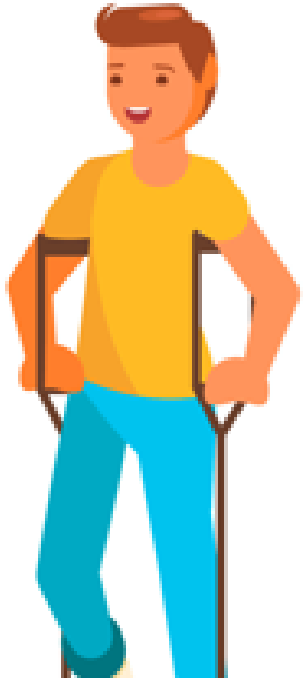
KTH VETENSKAP OCH KONST

Promobilia MoveAbility Lab





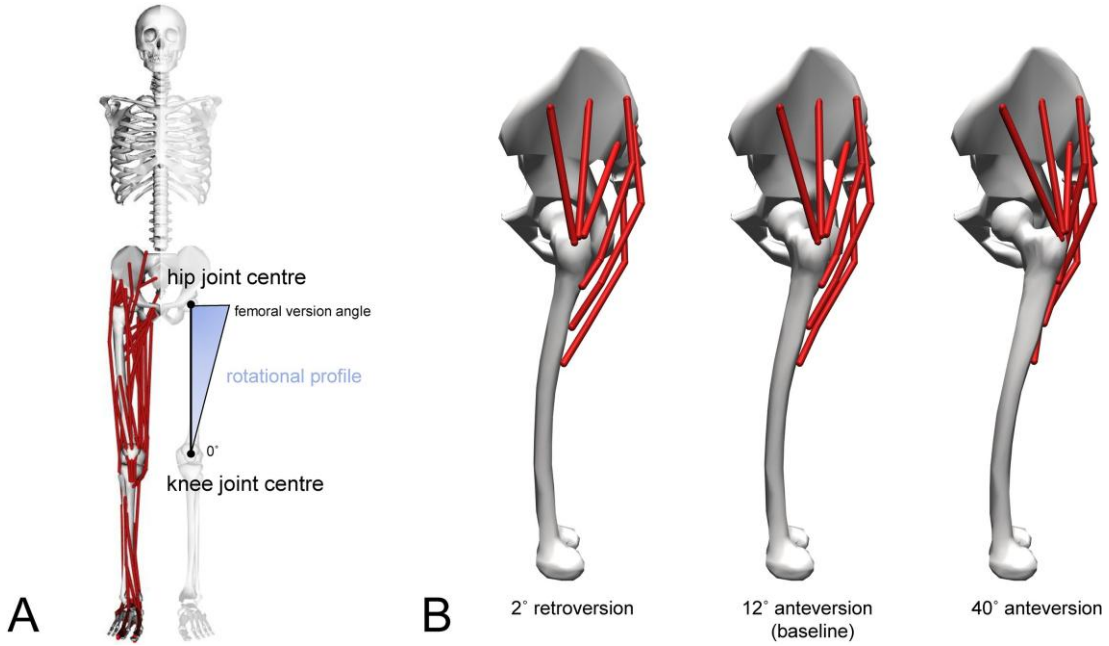




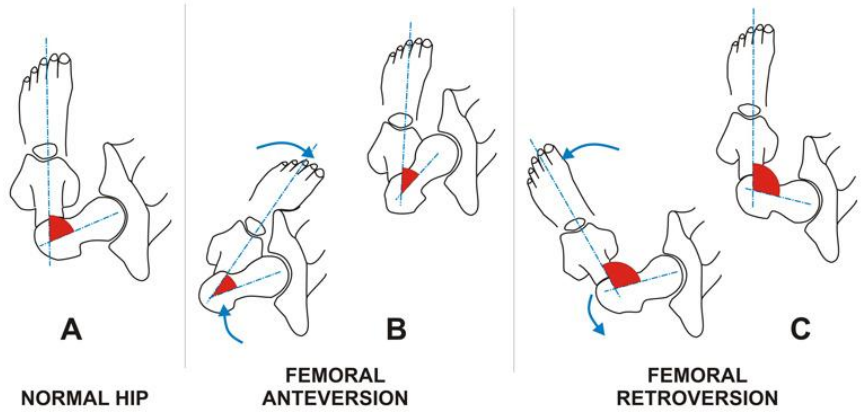
Why do we simulate human movement?



Better understanding of movement disorders



Modenese L., et al. Gait & Posture 2021

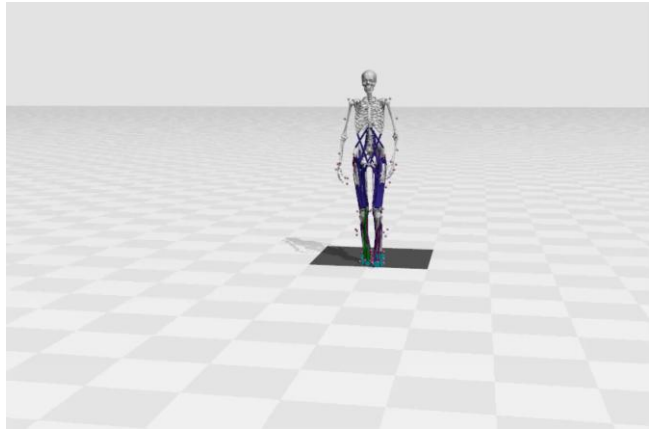


© 2011 Jake Pett, Stuart Pett

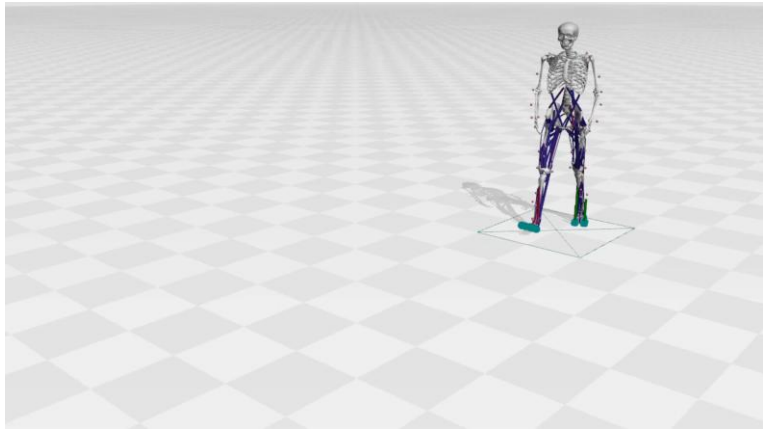
Compensatory gait strategies in individuals with muscle weakness



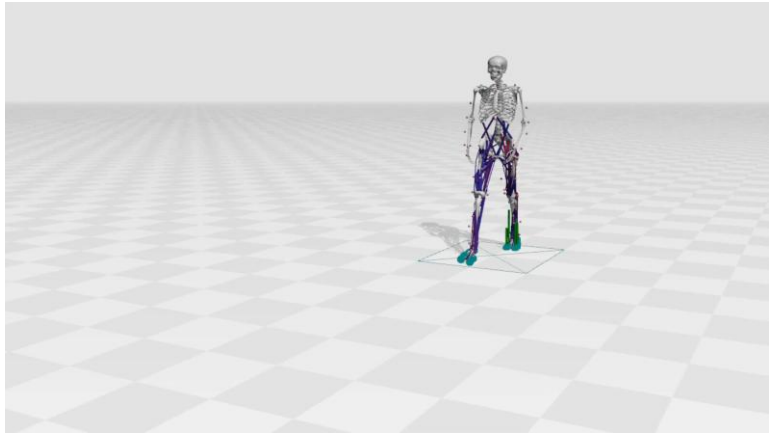
Minh Truong,
PhD student



Normal walking

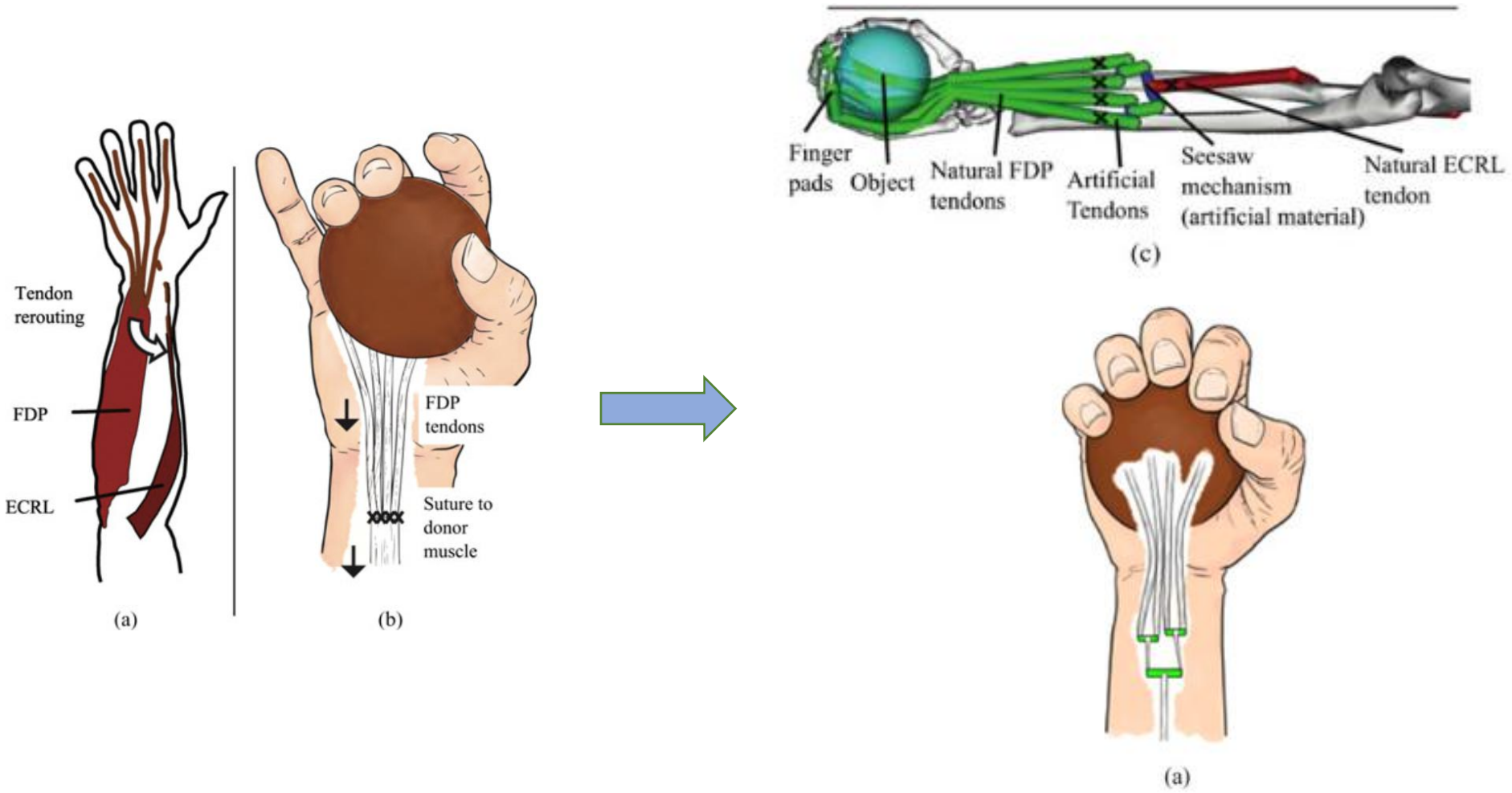


Weak hip abductors
(right side only)



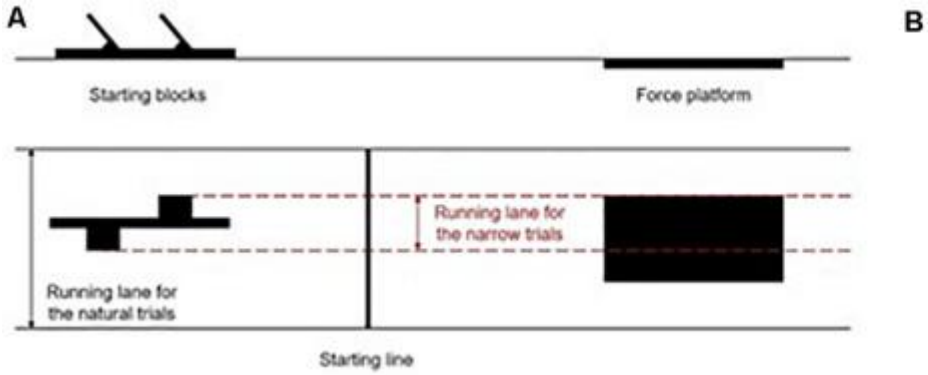
Weak ankle dorsiflexors
(right side only)

Predict outcomes from virtual surgery or interventions

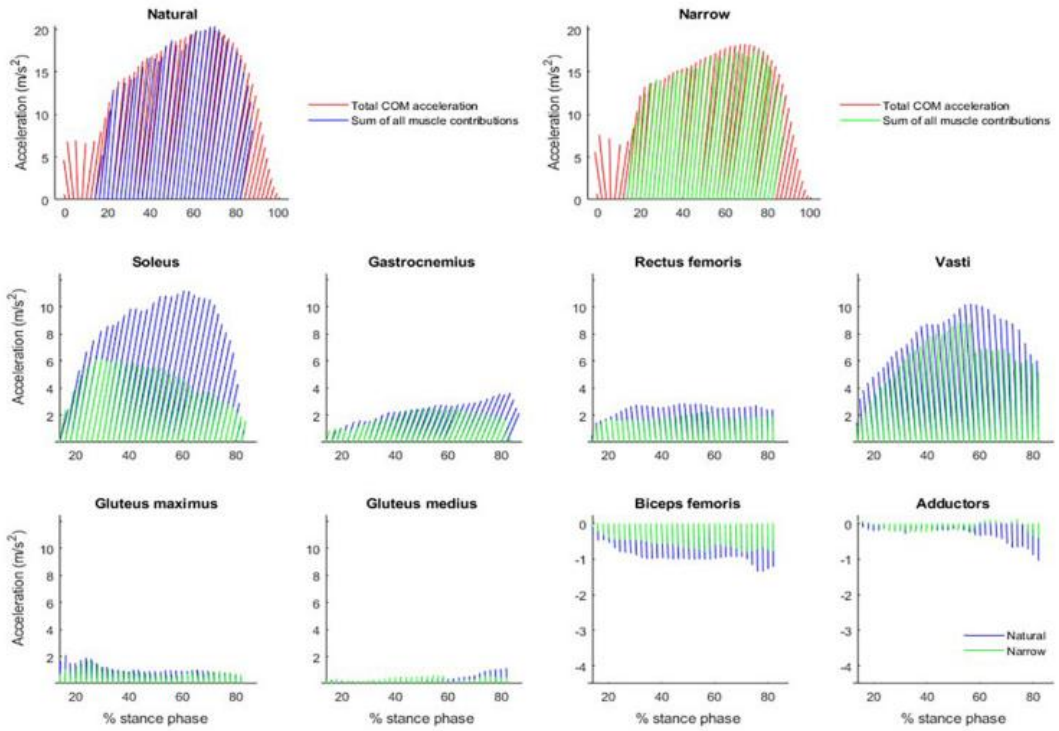


Homayouni T et al., IEEE transactions on biomedical engineering, 2015

Analyze and optimize sport performance







Neutral
Narrow

Wang R. et al., Frontiers in Bioeng Biotechnol. 2021

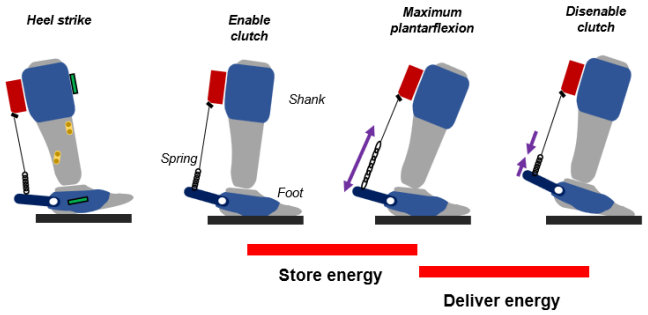
Developing neurorehabilitation technologies: human machine interaction



Israel Luis Pena, PostDoc

Promobilia MoveAbility Lab

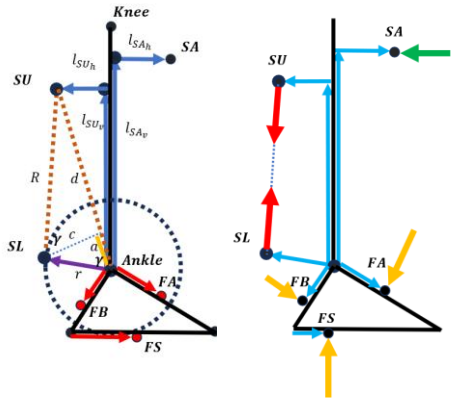
Most optimal **moments**



What is the “best” exoskeleton assistance?

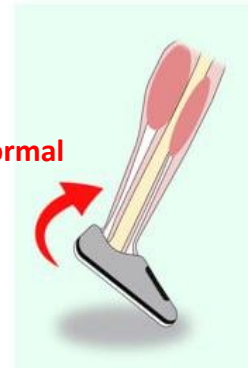
Interaction **forces** in the leg





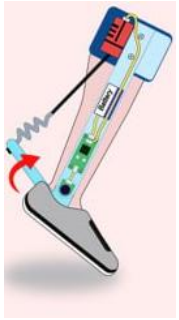
What configuration does it provide the lowest reaction forces in the leg?

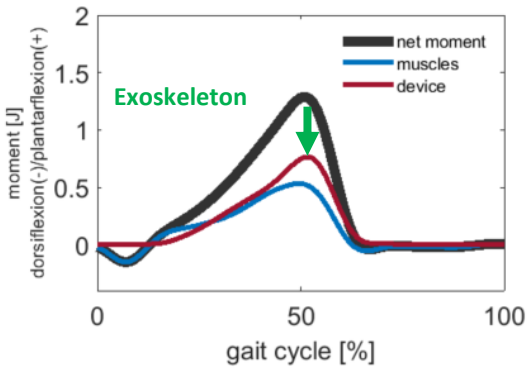
Muscle-tendon changes

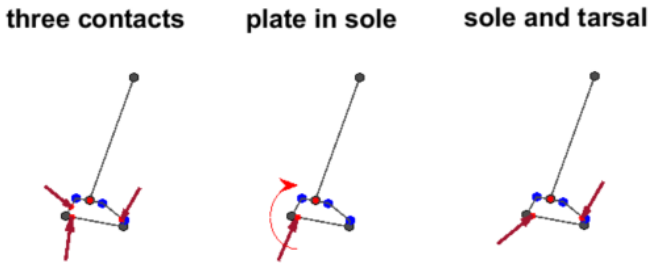




How muscle activity changes with exoskeleton moments?

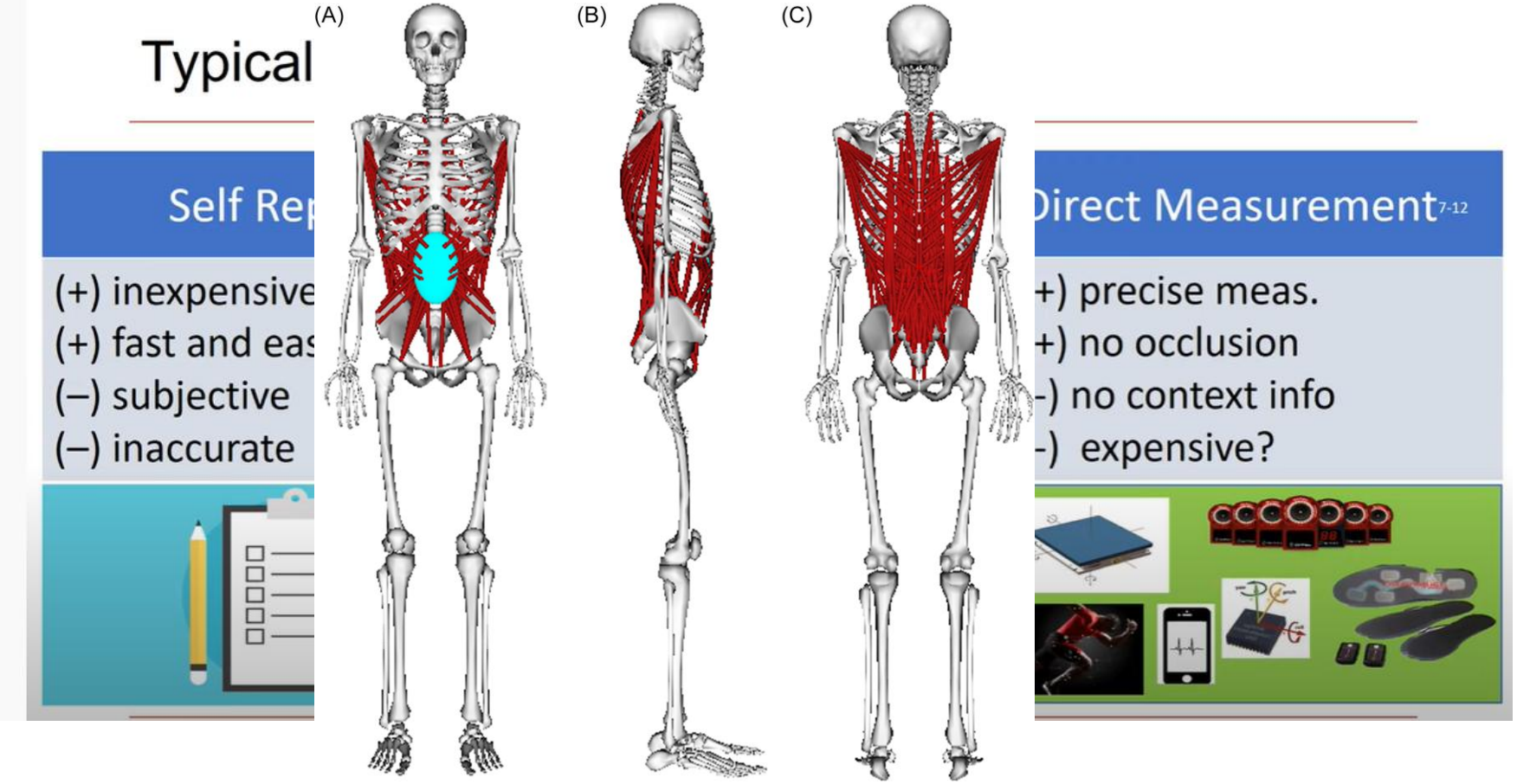






Find the optimal design parameters for exoskeletons

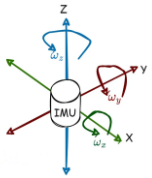
Quantitative Ergonomic Assessments



Beaucage-Gauvreau et al., Computer methods in biomechanics and biomedical engineering, 2019

Real-time full-chain biomechanical analysis platform with wearables

Promobilia MoveAbility Lab

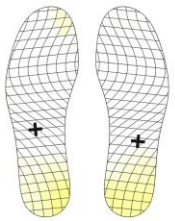


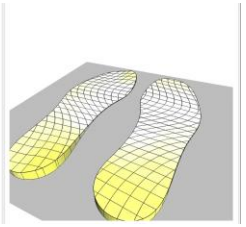


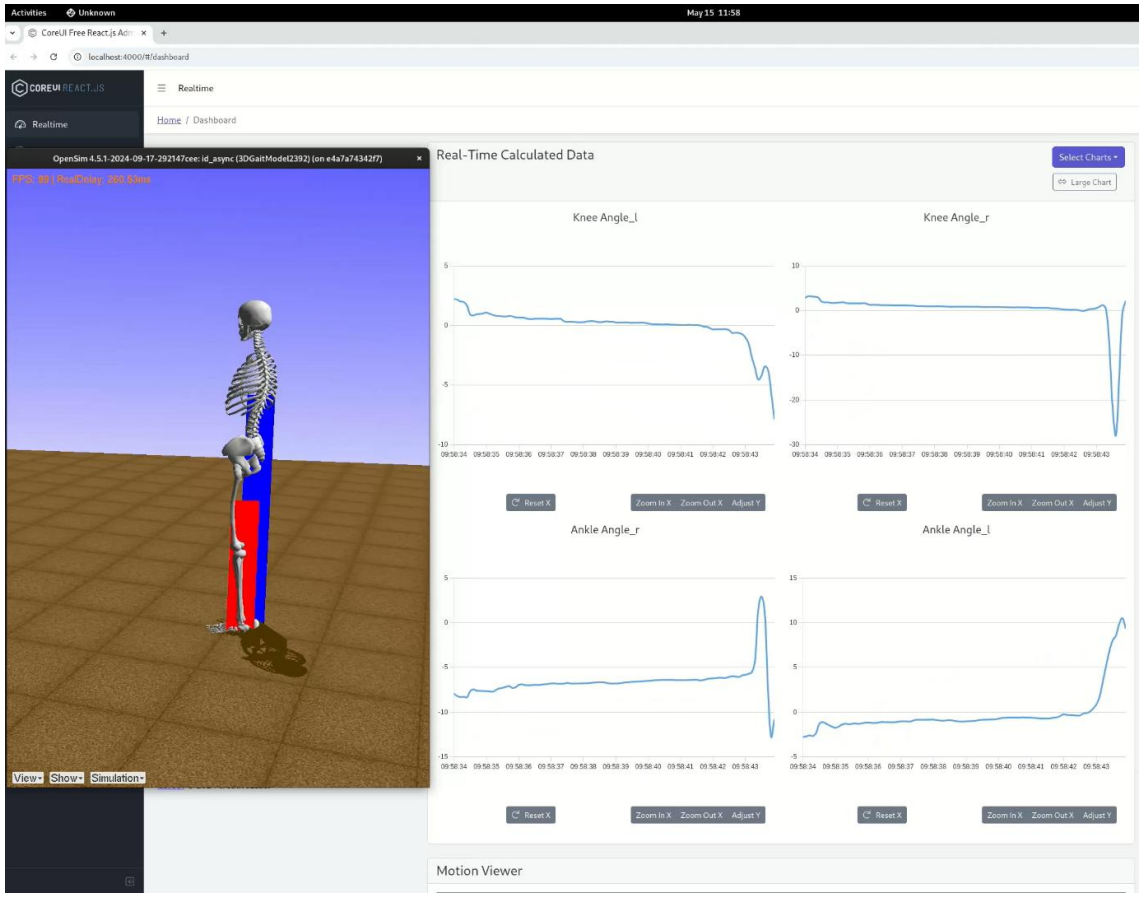
inertia measurement units (IMU)



Pressure insole











Zhaoyuan Wan
PhD student



Frederico Klein
PostDoc (alumni)

Why do we do simulation?

- Understand and better treatment of movement disorders
- Analyze and optimize sport performance
- Design ergonomically safe environment
- Design human-machine system
- ……..

See you in the first computer lab, 3rd Nov 10-12 in Christopher and Nils

Install OpenSim 4.5!!!